汇报提纲

一、研究背景

二、 MOR 一维孔道骨架掺杂思路

三、 Ce-MOR 亚埃级孔道调控及性能

四、结论与展望

一、研究背景

气体动力学直径

3 Å

C_2H_2(3.3Å)

CO_2(3.3Å)

O_2(3.5Å)

Ar(3.5Å)

N_2(3.64Å)

Kr(3.7Å)

CO(3.76Å)

CH_4(3.8Å)

4 Å

4 Å

Xe(4.1Å)

C_2H_4(4.2Å)

C_3H_4(4.2Å)

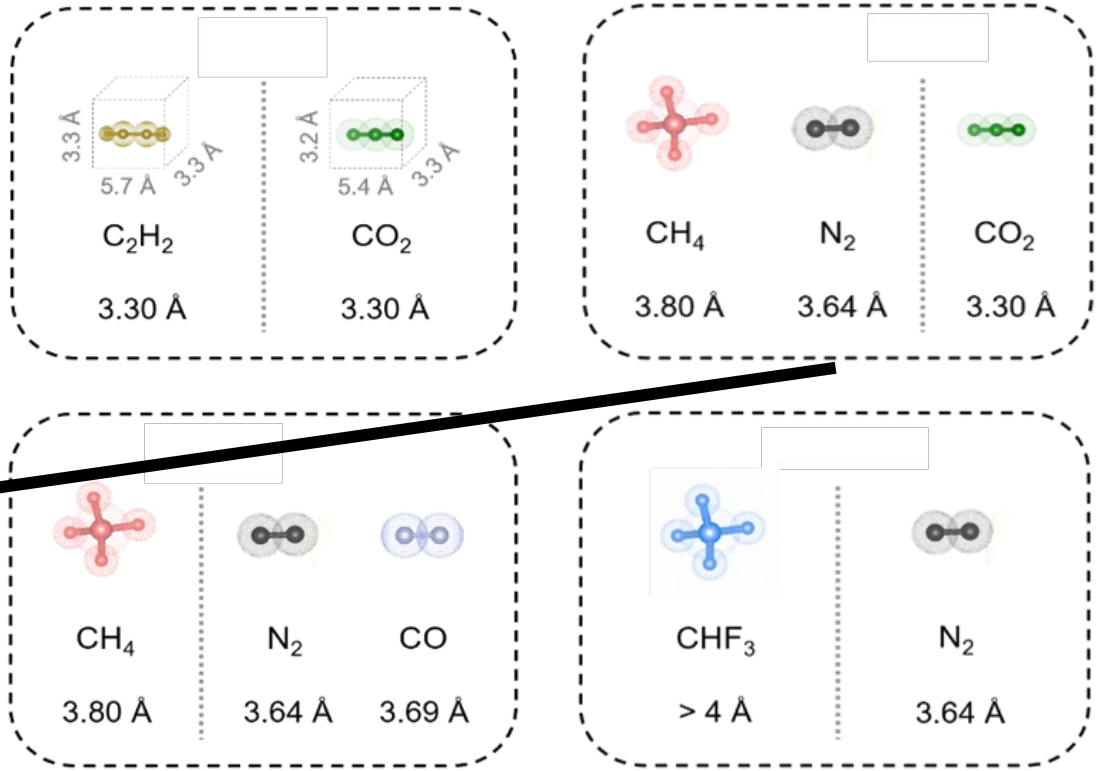
C_3H_8(4.3~5.1Å)

C_2H_6(4.4Å)

C_3H_6(4.7Å)

5 Å

相似气体体系分离

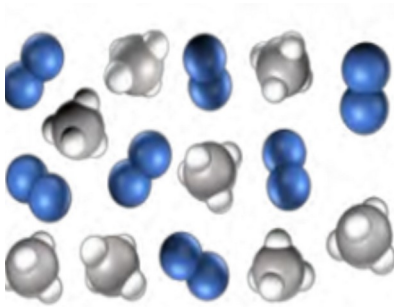


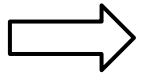
Adv. Mater. 2024, 36,

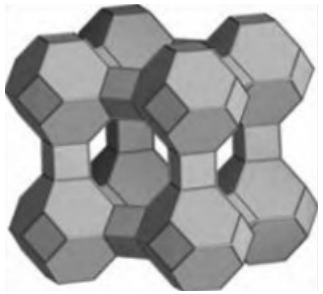
气体吸附分离

- 吸附分离：利用不同组分的热力学或动力学差异进行分离，实现混合气体分离提纯

  优点：无相变、能耗低；操作简单、便于集成化控制



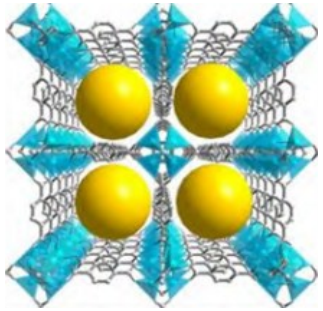




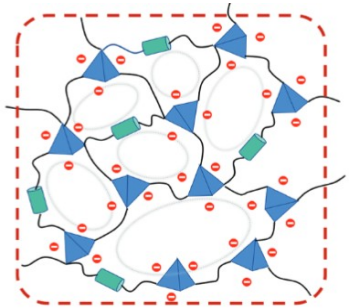
Zeolites



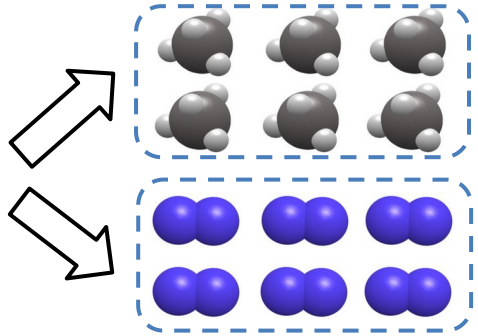
Carbon based



MOF

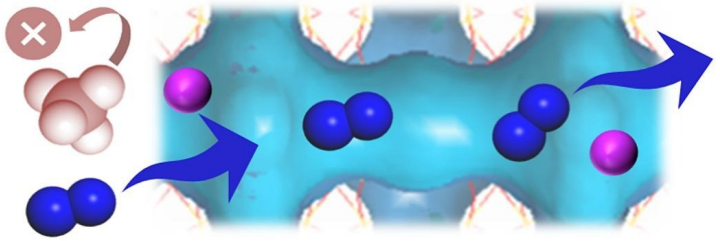


POF



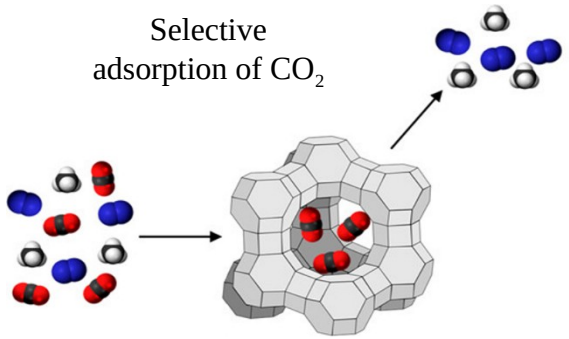
吸附分离机理

尺寸筛分效应



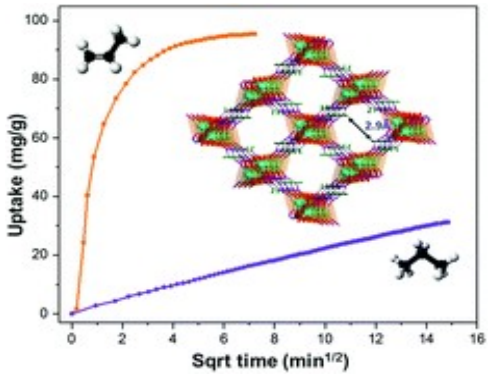
Chem. Eng. J., 2024, 495, 153630

作用力差异效应



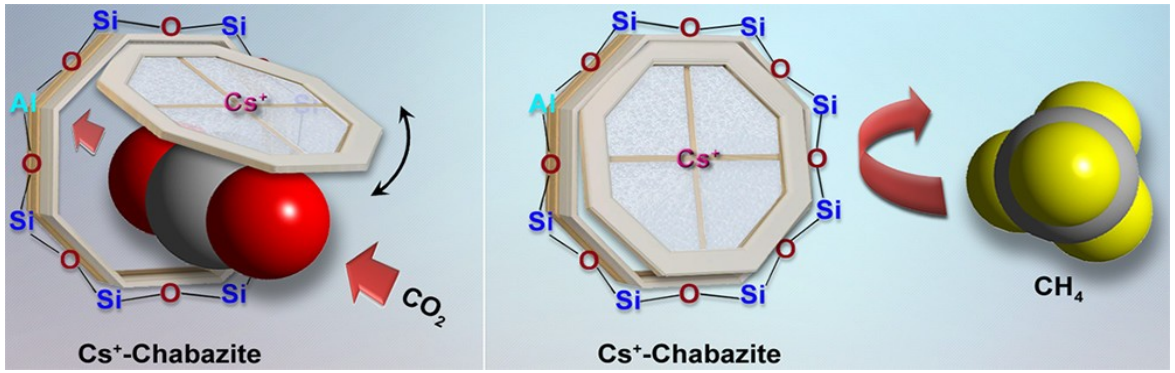
ACS Appl. Energy Mater. 2023, 6, 2634−2656

扩散速率差异效应



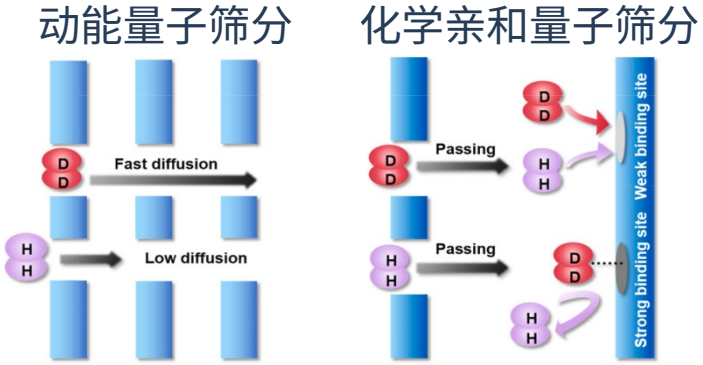
Chem. Commun., 2017, 53, 9332-9335

合页门（trapdoor）效应



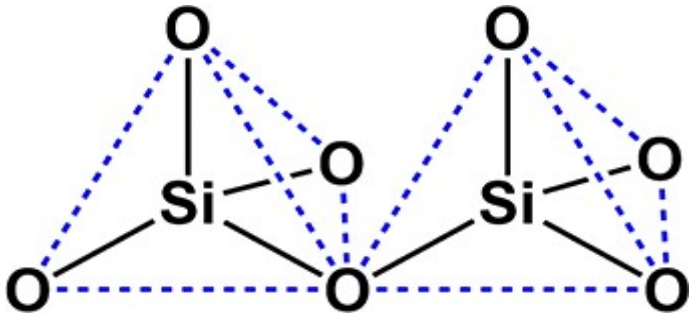
J. Am. Chem. Soc., 2012, 134, 46, 19246–19253

量子筛分效应

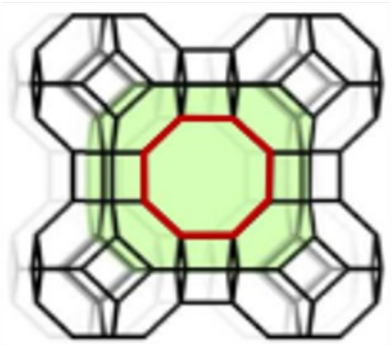


Chem. Phys. Lett., 1995, 232(4): 379-82

沸石分子筛

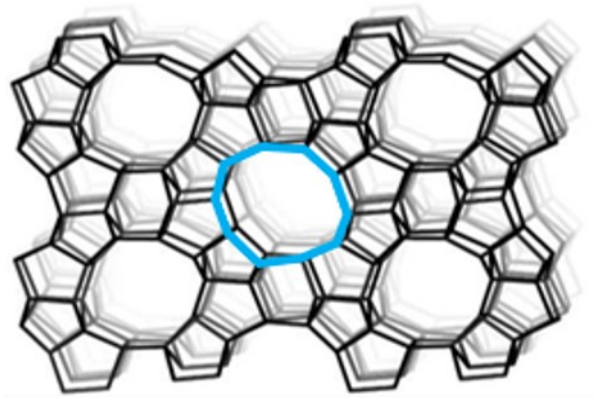


TO4 （ T=Si ， AI 等）四面体共顶点连接而成规则孔道结构



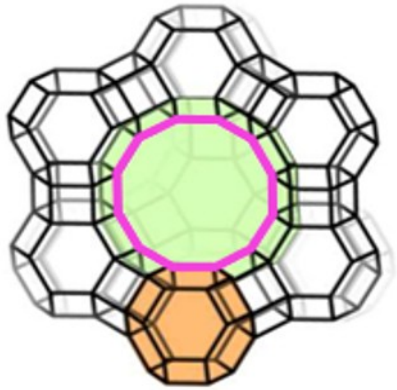
LTA

小孔
8 元环（ ~4 Å ）



MFI

中孔
10 元环（ 5~6 Å ）



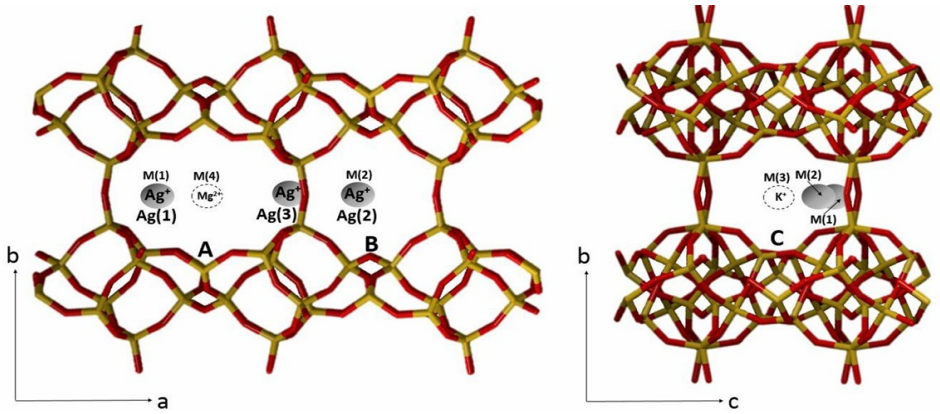
FAU

大孔
12 元环（ 6~8 Å ）

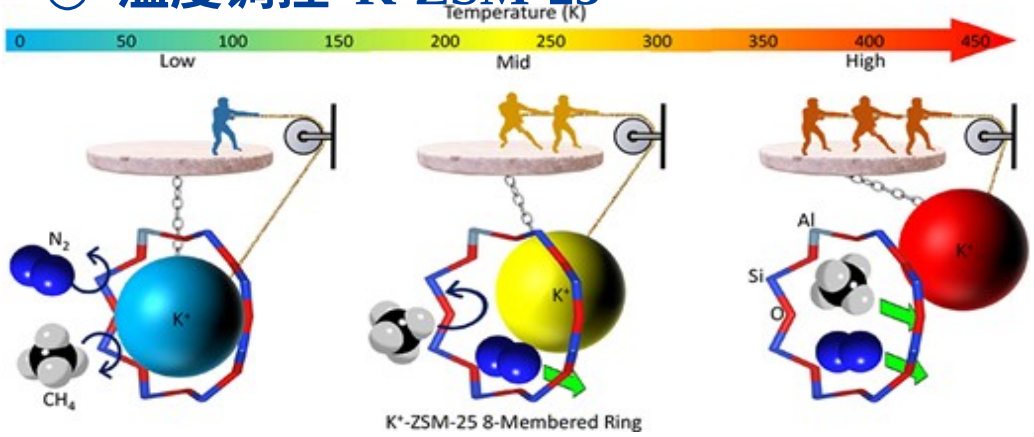
Chem. Rev. (2022) 122 (24): 17647–17695. *ACS Appl. Energy Mater.* (2023) 6 (5): 2634–2656.

分子筛孔径调控策略

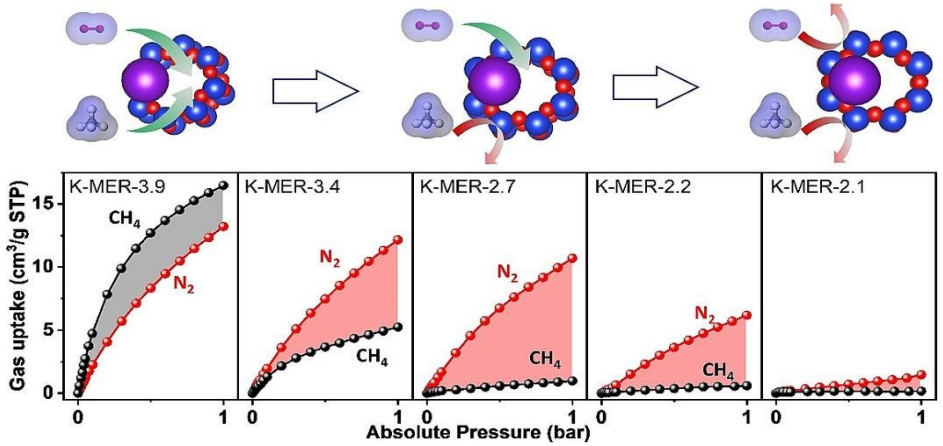
◎ 离子交换 Ag-clinoptilolite



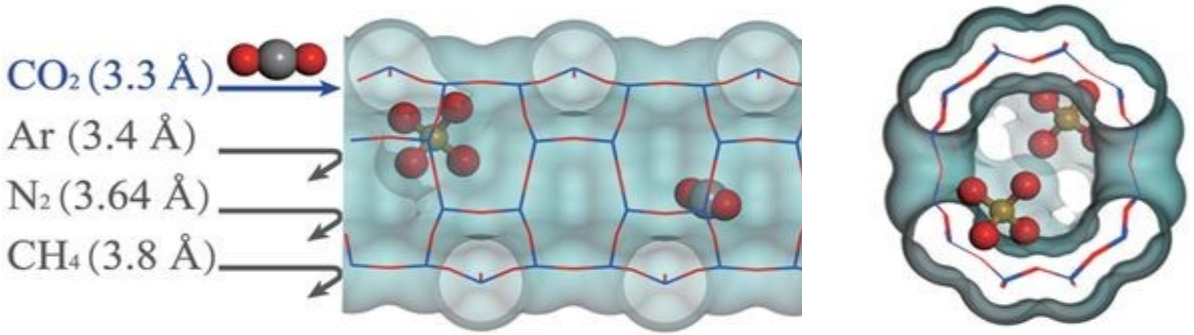
◎ 温度调控 K-ZSM-25



◎ 硅铝比调控 K-MER



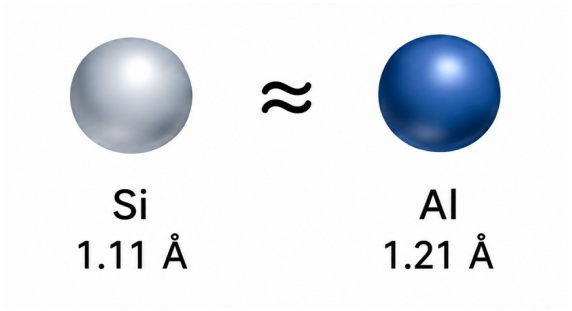
◎ 骨架外物种堵孔 Fe-MOR



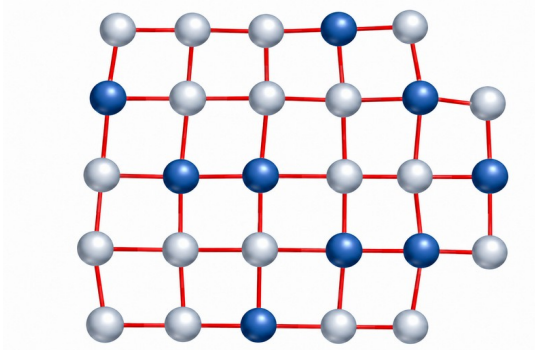
挑战：实现分子筛亚埃级孔道尺寸连续调控

Microporous Mesoporous Mater., 2019, 281: 123. *J. Am. Chem. Soc.*, 2021, 143: 15195. *Sep. Purif. Technol.*, 2024, 339: 126601. *Science*, 2021, 373: 315.

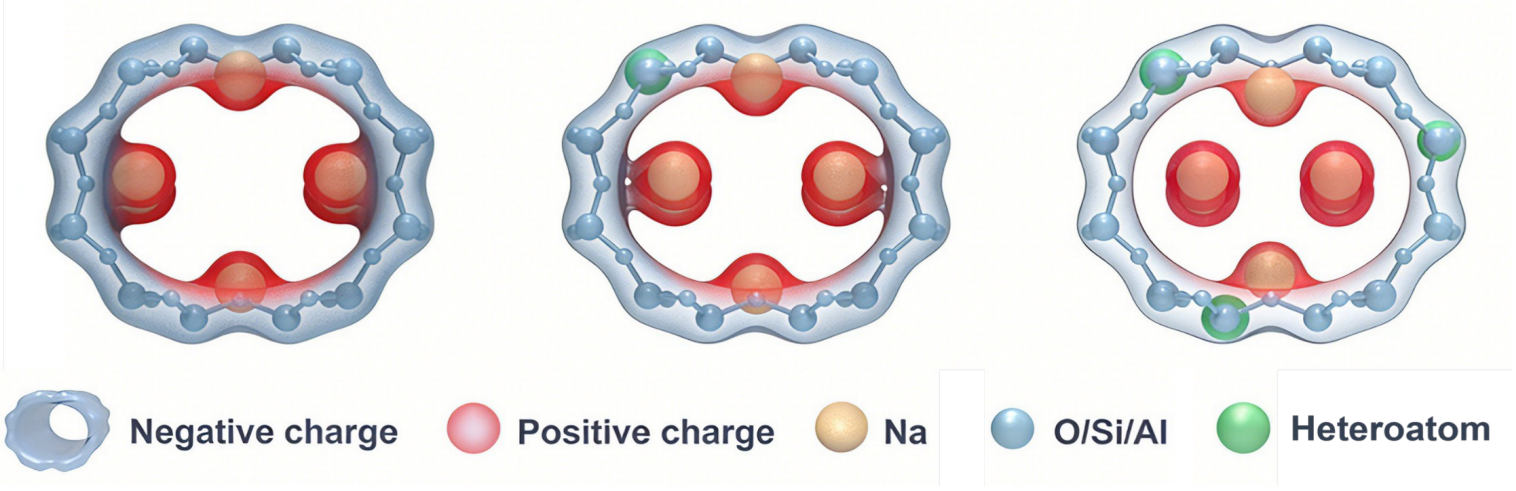
二、 MOR 一维孔道骨架掺杂思路



Si 与 Al 原子半径相似



Al 原子分布存在随机性



❑ 调控骨架电荷诱导阳离子位置可控迁移，有望实现孔径亚埃级精度调控

三、 Ce 掺杂 MOR 结构表征

❑ Ce 掺杂未改变 MOR 拓扑结构

◎ Ce 离子交换与原位掺杂 MOR 差异明显

◎ 掺杂前后铝均以四配位形式存在

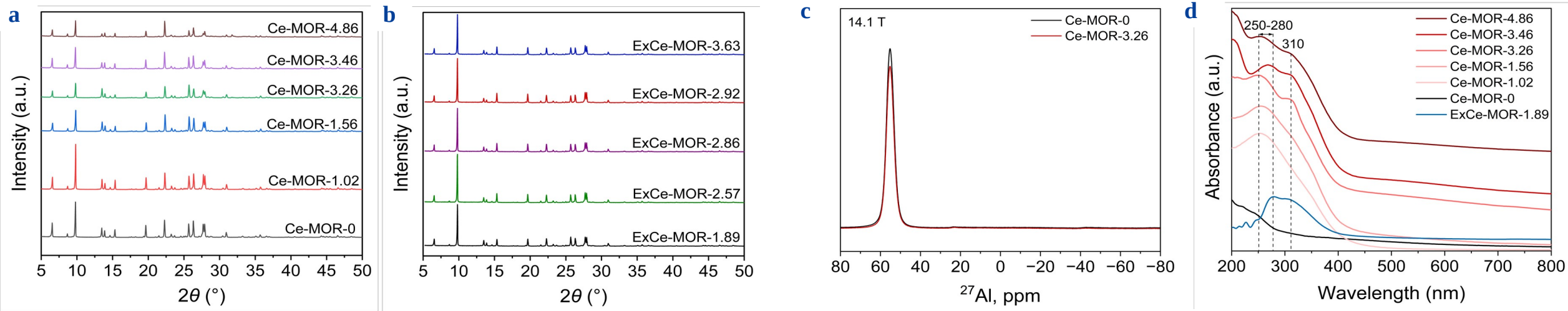
◎ 未见非骨架铝物种

❑ Ce 存在形式取决于引入方式与掺杂量

◎ 离子交换：骨架外阳离子

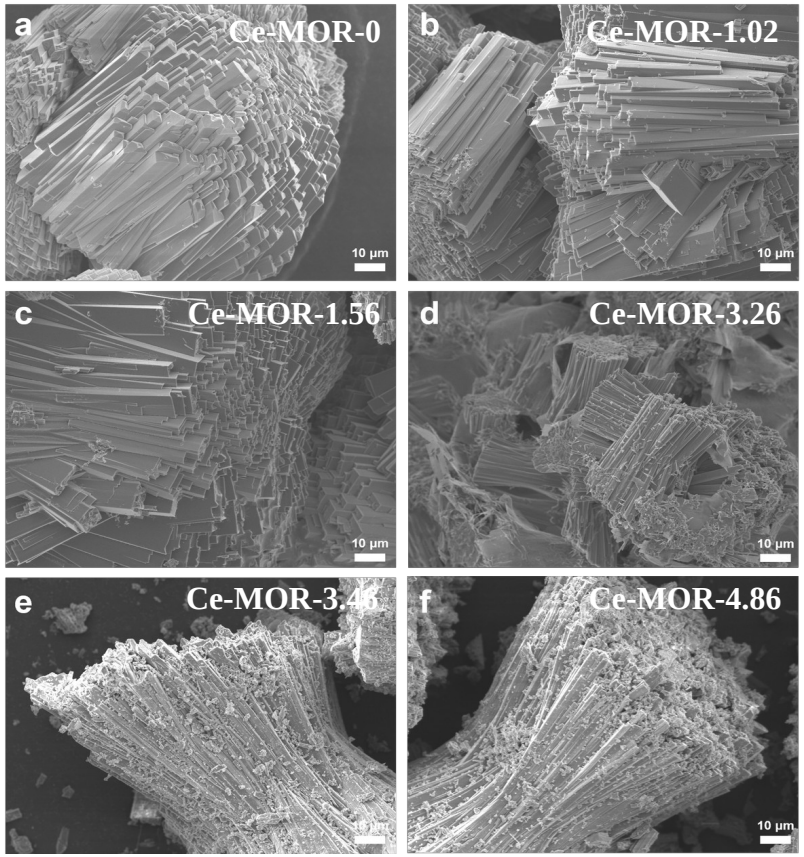
◎ 原位掺杂： Ce≤1.56%：骨架原子；

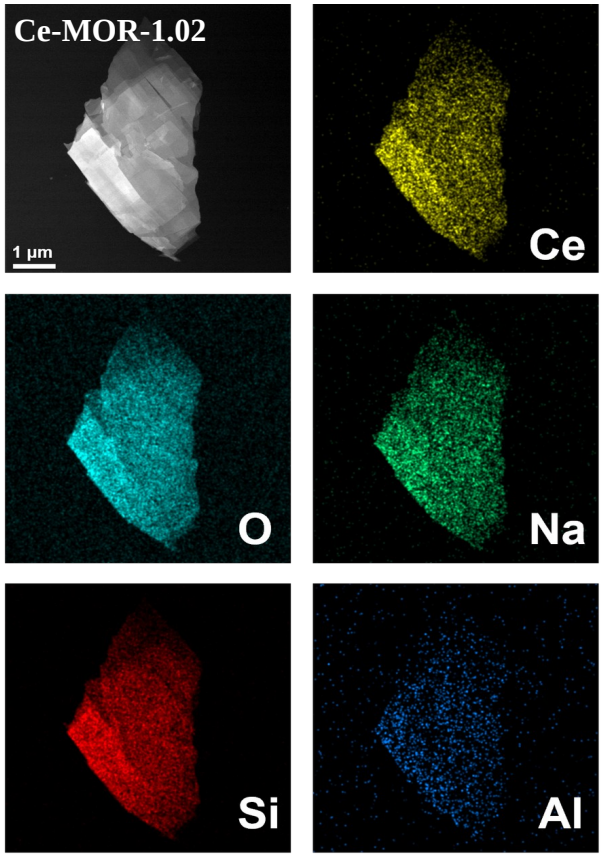
Ce ≥3.26%：出现骨架外 Ce 物种

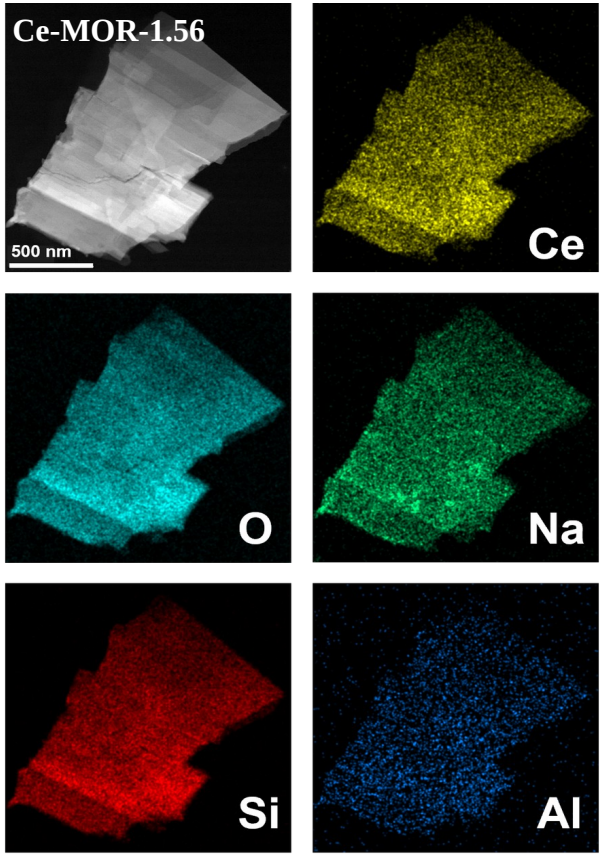


Ce 掺杂及其元素分布

❑ Ce 掺杂量存在阈值



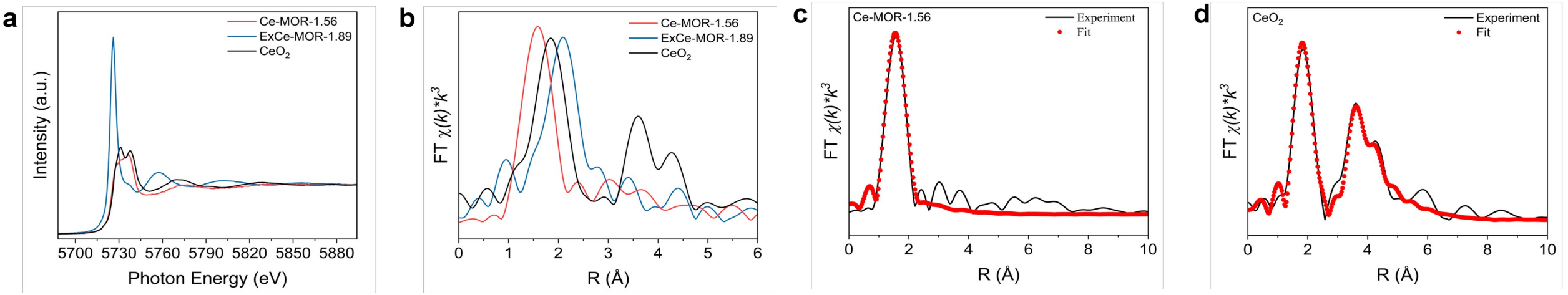




Revised

杂原子 Ce 的存在状态

- Ce 掺杂：边前峰与 CeO_2 类似，配位数 4 ， Ce-O 键长 2.13 Å
- Ce 例子交换：与 $Ce(NO_3)_3$ 一致（ Ce^{3+} ）

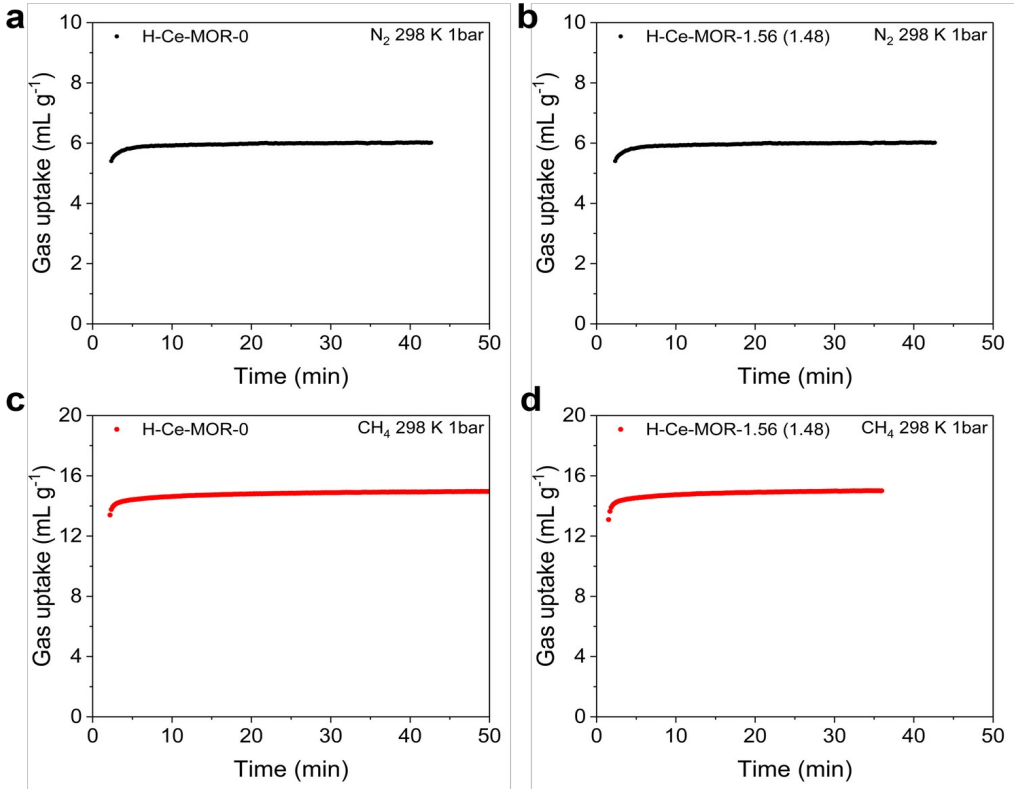


掺杂 Ce 以 Ce(IV) 形式进入骨架，离子交换 Ce 以 Ce(III) 形式存在于孔道

Revised

气体吸附动力学分析

- 质子交换可选择性去除孔道内 Na^+ ，而 Ce 含量保持不变
- 去除 Na^+ 后， CH_4 和 N_2 的吸附速率显著加快，短时间内即可达到平衡，扩散阻力大幅降低
- 证实 Ce 掺杂是通过诱导 Na^+ 迁移来改变有效孔径，进而调控气体扩散行为

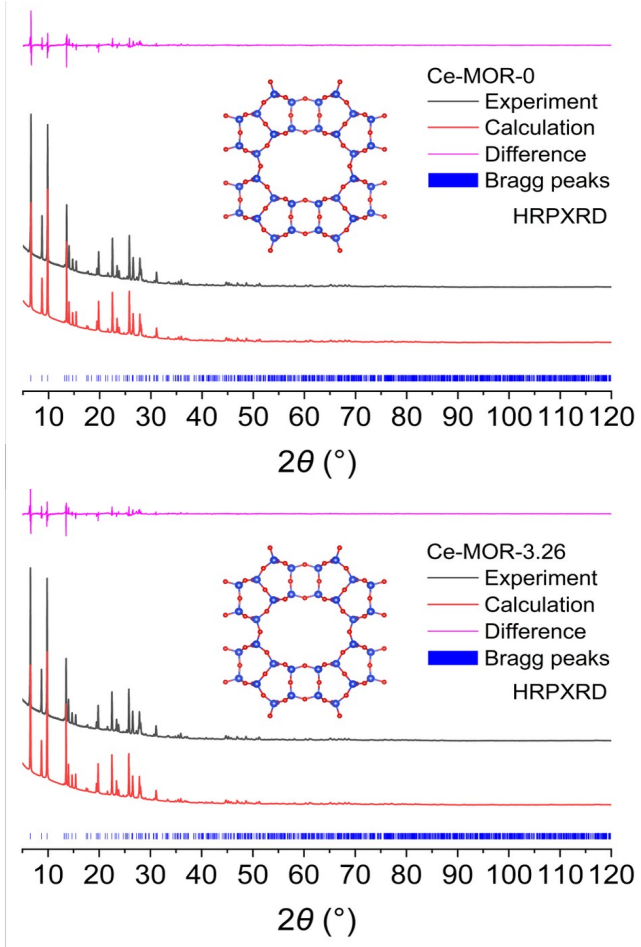


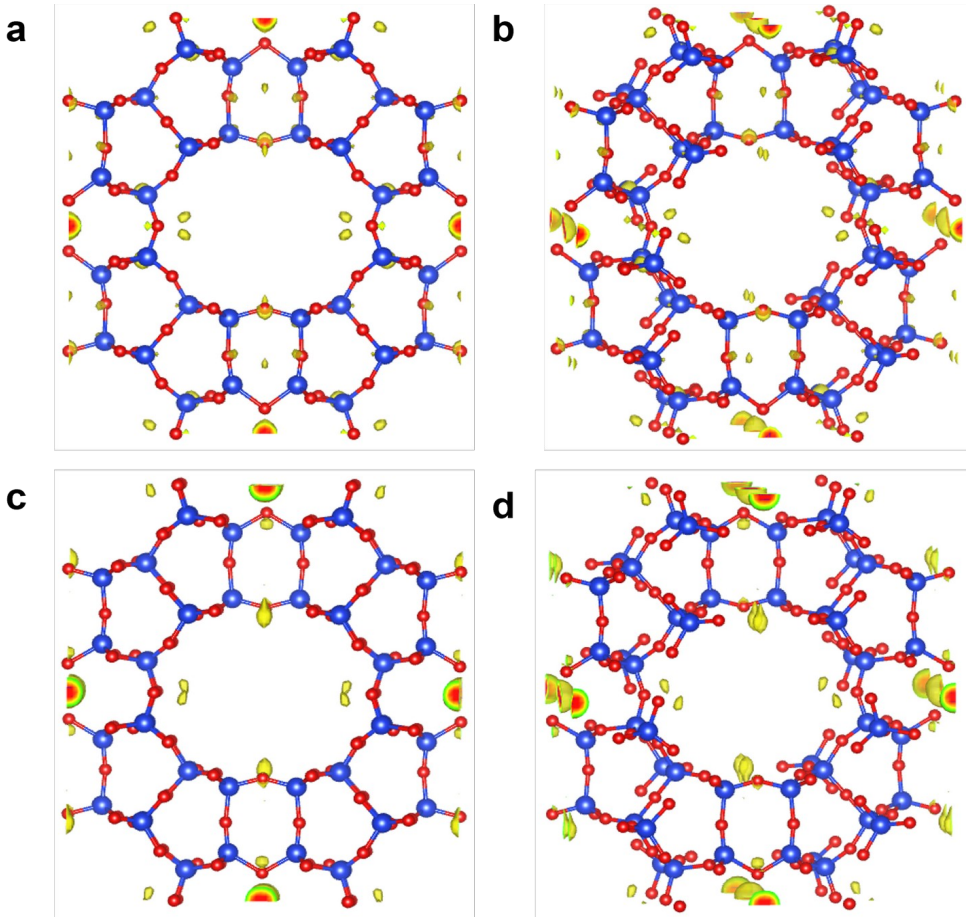
样品	质量分数			
	Si (wt%)	Al (wt%)	Na (wt%)	Ce (wt%)
H-Ce-MOR-0	83.71	15.87	0.43	0.00
H-Ce-MOR-1.56 (1.48)	83.15	15.07	0.29	1.48

Revised

阳离子位点解析

- Ce 掺杂前后阳离子分布相似，但 12MR 内位置发生明显变化

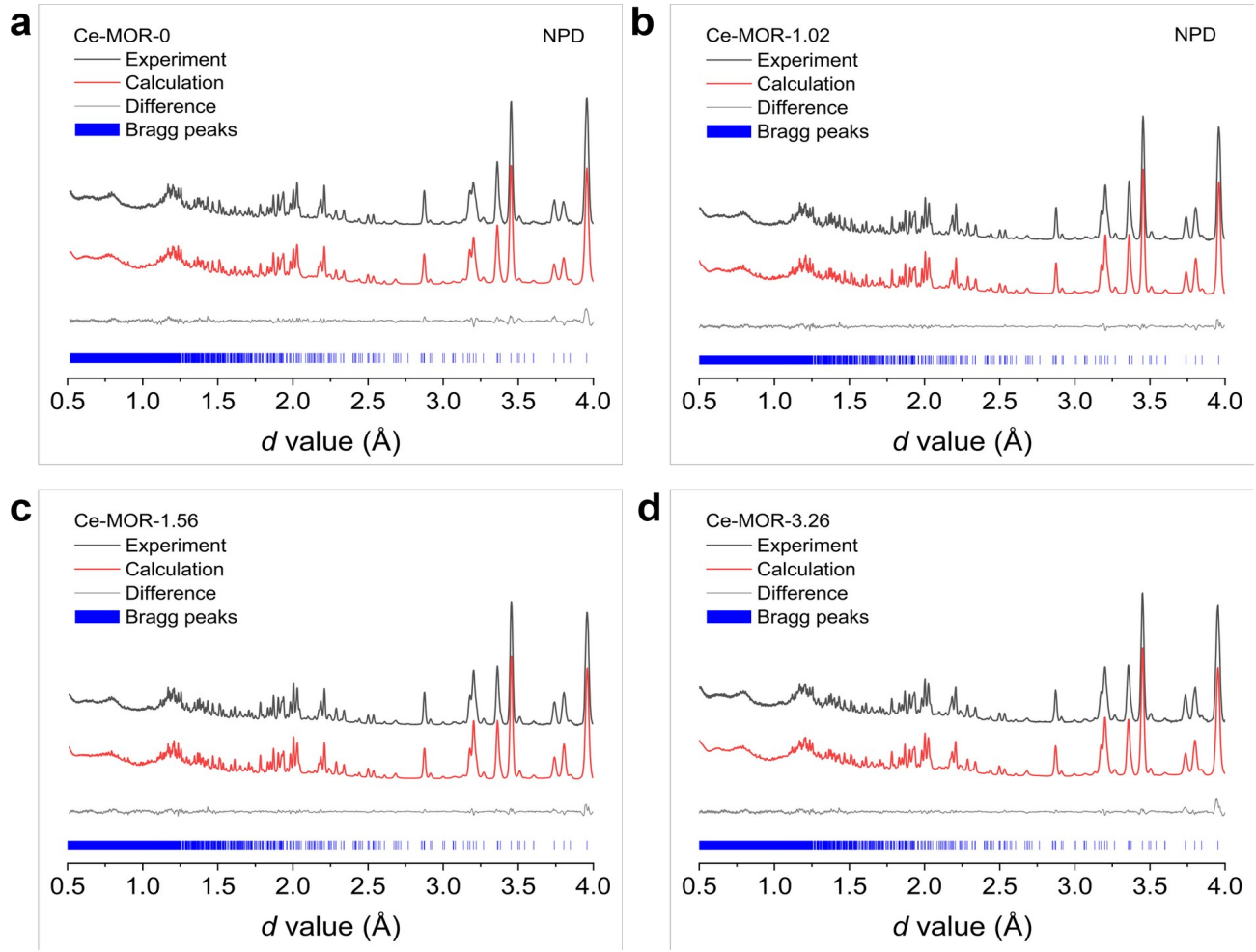


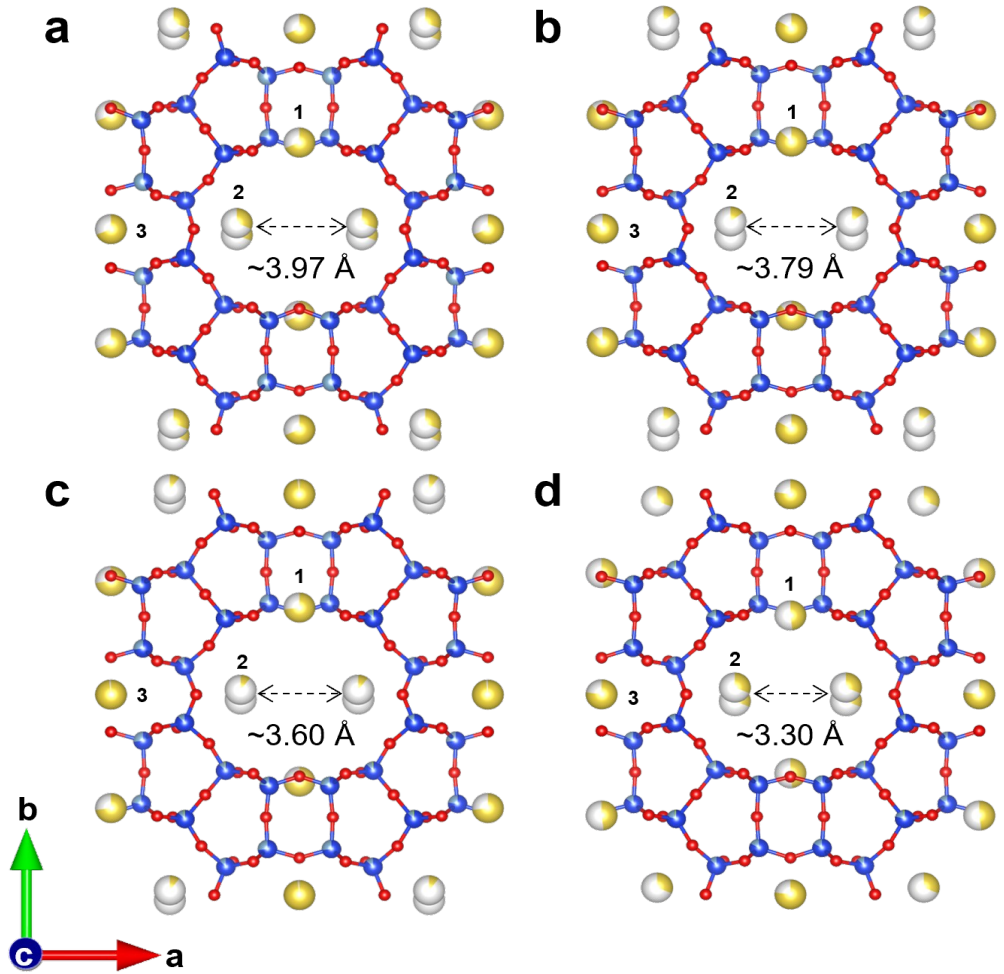


Revised

Ce-MOR 的结构解析

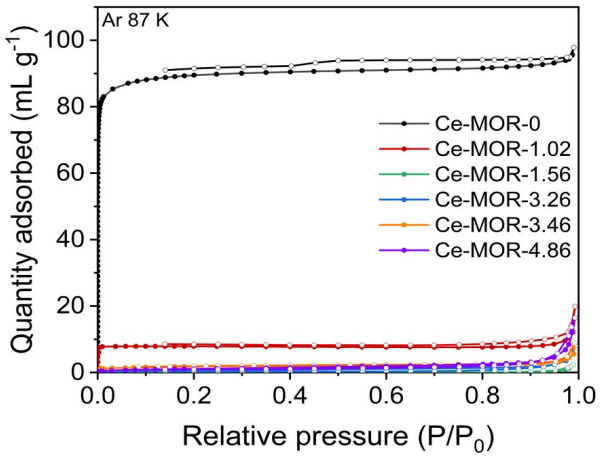
- Ce 掺杂使 12MR 内 Na–Na 间距由 3.97 Å 逐步缩短至 3.30 Å，实现孔道可控收缩

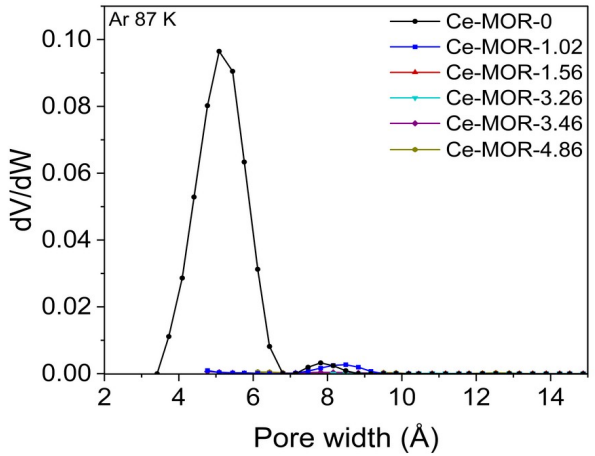


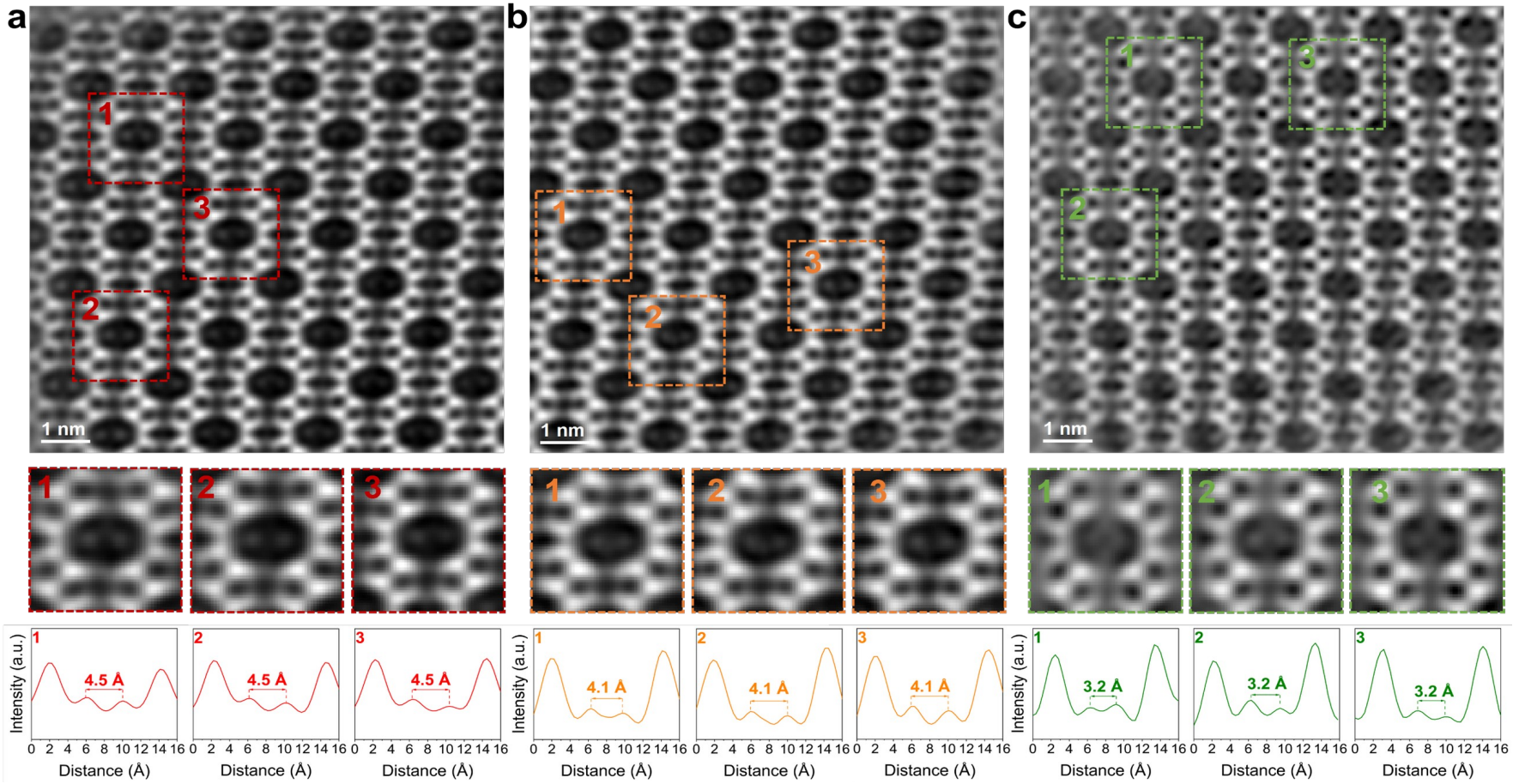


Revised

Ce-MOR 孔径及球差电镜分析





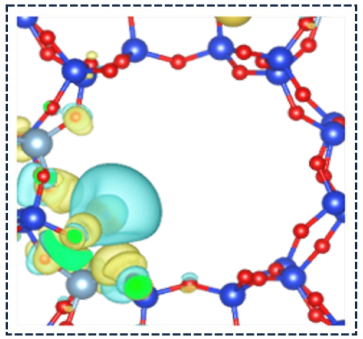


Revised

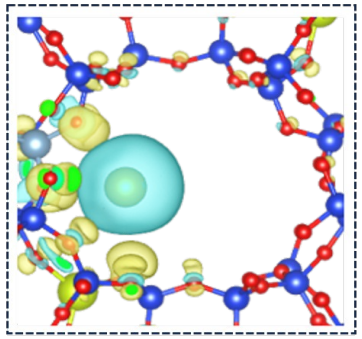
电荷诱导阳离子迁移机制

❑ 杂原子掺杂改变骨架电荷分布，驱动 Na^+ 向孔道中心迁移

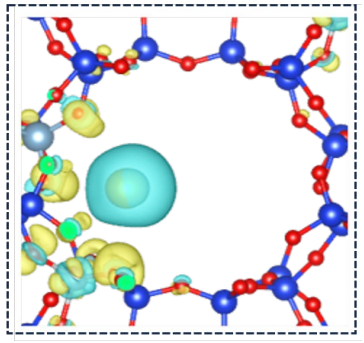
纯 MOR



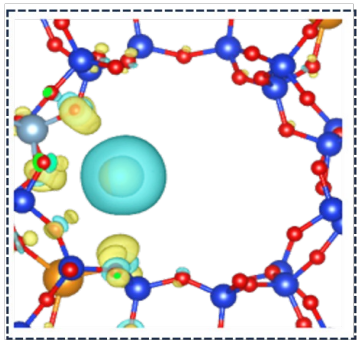
Ce



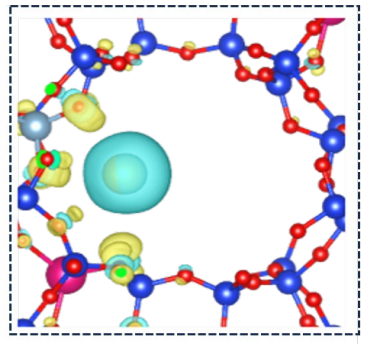
Ni



Nd



Sm

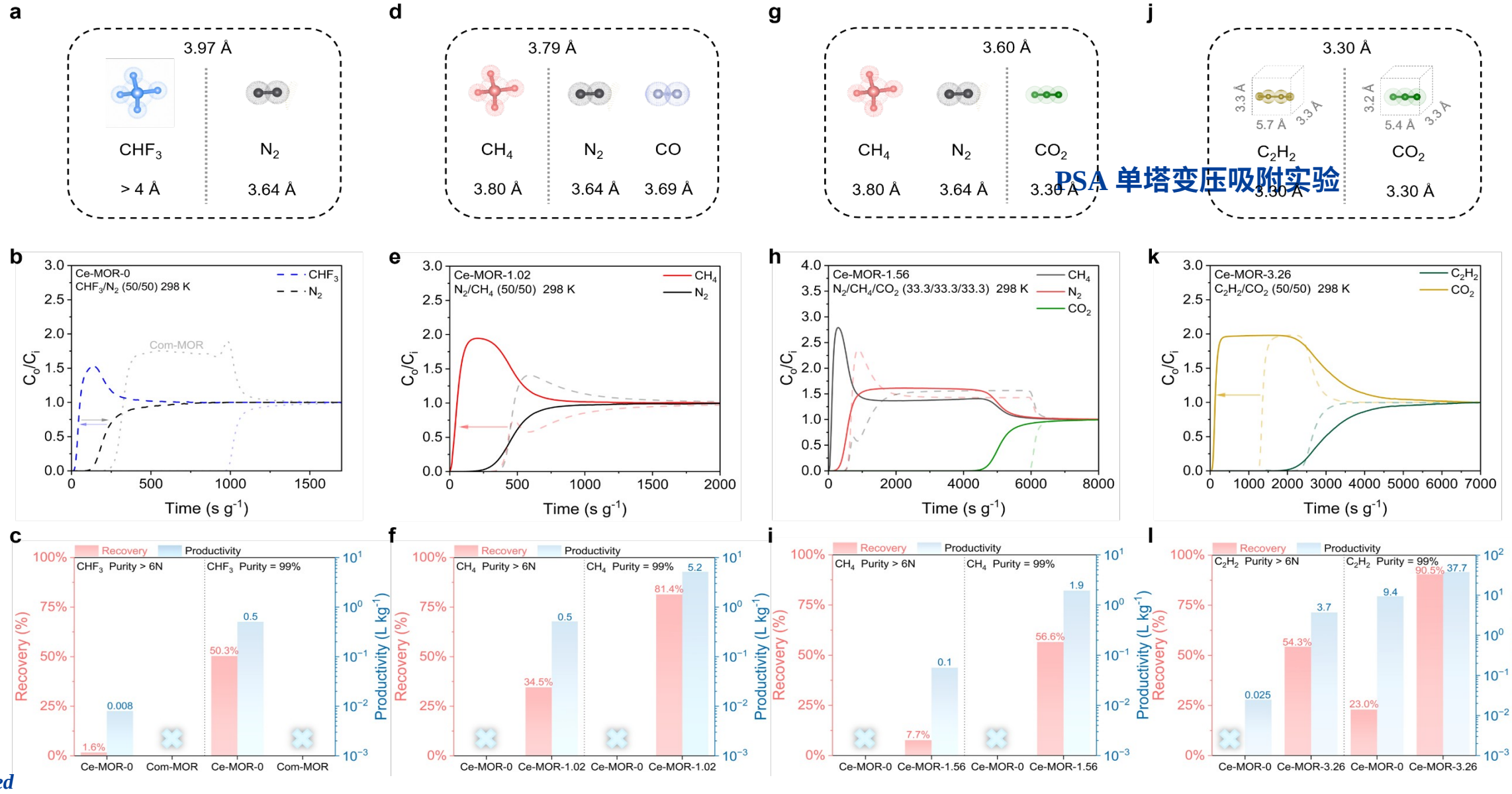


	掺 Ce 体系	未掺杂 Ce 体系	Δq (e)
O1	-1.3831	-1.5991	0.2160
O2	-1.3664	-1.6191	0.2527
O3	-1.3772	-1.5920	0.2148
O4	-1.3510	-1.5995	0.2485
Total	-5.4777	-6.4097	0.9320

Revised

相似气体分子体系分离性能

PSA 单塔变压吸附实验



Revised